

Sign of permutation.

Let a polynomial over  $\{\pm 1\}$  with distinct  $n$ -variables  $x_1, \dots, x_n$  as

$$\Delta \stackrel{\text{def}}{=} \prod_{1 \leq i < j \leq n} (x_i - x_j)$$

For each  $\sigma \in S_n$ , let  $\sigma$  act on  $\Delta$  by permuting:

$$\sigma(\Delta) = \prod_{1 \leq i < j \leq n} (x_{\sigma(i)} - x_{\sigma(j)})$$

Since  $\sigma$  is bijective, there are two possibilities:

$$\sigma(\Delta) = \pm \Delta.$$

Now, let  $\text{sign}: S_n \rightarrow \{\pm 1\}: \sigma \mapsto \begin{cases} 1 & \text{if } \sigma(\Delta) = \Delta \\ -1 & \text{if } \sigma(\Delta) = -\Delta \end{cases}$ .

Prop. Sign is a surjective homomorphism.

Proof. Given  $\sigma, \tau \in S_n$ ,

$$(\tau \circ \sigma)(\Delta) = \prod_{1 \leq i < j \leq n} (x_{(\tau \circ \sigma)(i)} - x_{(\tau \circ \sigma)(j)})$$

Suppose  $\sigma(\Delta)$  has exactly  $k$  factors of  $(x_i - x_j)$  with  $j > i$ . That is,  $\text{sign } \sigma = (-1)^k$ .

$$\text{So, } \sigma(\Delta) = \prod_{1 \leq i < j \leq n} (x_{\sigma(i)} - x_{\sigma(j)}) = \text{sign } \sigma \cdot \prod_{1 \leq p < q \leq n} (x_p - x_q)$$

Therefore,  $(\tau \circ \sigma)(\Delta) = \text{sign } \sigma \cdot \prod_{1 \leq p < q \leq n} (x_{\tau(p)} - x_{\tau(q)})$ , it follows

$$\text{sign } (\tau \circ \sigma) = \text{sign } \sigma \cdot \text{sign } \tau.$$

Surjectiveness: clearly,  $\text{sign}(id) = 1$ . And,  $\text{sign}((12)) = -1$ , so clear.

Prop.  $\ker \text{sign} = A_n$ .

Prop.  $m$ -cycle is odd iff  $m$  is even

Prop. a permutation  $\sigma$  is odd iff number of cycles of even length is odd, in decomposition.

[Prop.  $S_n$  is generated by  $\{(i \ i+1) \mid 1 \leq i \leq n-1\}$ .

Proof. Given  $j \in \{2, \dots, n-1\}$ ,  
 $(1 \ 2 \ 3 \ \dots \ n) = (1 \ n)(1 \ n-1) \dots (1 \ 3)(1 \ 2)$   
 $(1 \ 2)(2 \ 3) \dots (j-1 \ j) = (1 \ 2 \ 3 \ \dots \ j)$

So, the generated set contains  $(1 \ 2 \ \dots \ j)$  for all  $2 \leq j \leq n$ .

Moreover, observe that

$$(2 \ 3)(1 \ 2)(2 \ 3)^{-1} = (2 \ 3)(1 \ 2 \ 3) = (1 \ 3)$$

$$(3 \ 4)(1 \ 3)(3 \ 4) = (3 \ 4)(1 \ 3 \ 4) = (1 \ 4)$$

$$\vdots$$

$$(j \ j+1)(1 \ j)(j \ j+1) = (1 \ j)$$

Now, for any  $1 \leq j < k \leq n$ ,

$$(j \ k) = (1 \ k)(1 \ j)$$

can be generated by the given set. All permutations can be represented by product of transpositions, so  $S_n = \langle (i \ i+1) \mid 1 \leq i \leq n-1 \rangle$

[Corollary,  $S_n = \langle (1 \ 2), (1 \ 2 \ \dots \ n) \rangle$

Proof. observe that

$$(1 \ 2 \ \dots \ n)(1 \ 2)(1 \ 2 \ \dots \ n)^{-1} = (1 \ 2 \ \dots \ n)(1 \ 2)(n \ n-1 \ \dots \ 1) = (2 \ 3)$$

$$(1 \ 2 \ \dots \ n)(2 \ 3)(1 \ 2 \ \dots \ n)^{-1} = (3 \ 4)$$

$\vdots$

So,  $\langle (1 \ 2), (1 \ 2 \ \dots \ n) \rangle$  contains  $(i \ i+1)$  for all  $1 \leq i \leq n-1$ , proved

+ Note that

$$(a_1 \ a_2 \ \dots \ a_n)$$

$$= (a_1 \ a_n)(a_1 \ a_2 \ \dots \ a_{n-1})$$

$$= (a_1 \ a_n)(a_1 \ a_{n-1})(a_1 \ a_2 \ \dots \ a_{n-2})$$

$$\vdots$$

$$= (a_1 \ a_n)(a_1 \ a_{n-1}) \dots (a_1 \ a_2)$$

Thm. For a prime number  $p$ ,  $S_p$  is generated by any one transposition and  $p$ -cycle.  
 Proof. WLOG, let the one transposition is  $(1\ x)$  and  $p$ -cycle is  $(1\ \dots\ p)$ .  
 where  $x \in \{2, \dots, p\}$ . And, since  $p$  is prime number,

$$(1\ 2\ \dots\ p)^k \quad (1 \leq k \leq p-1)$$

are all  $p$ -cycles. And, take  $k = x-1$ , then

$$(1\ 2\ \dots\ p)^k = (1\ x\ \dots)$$

$$\text{So } \langle (1\ x), (1\ \dots\ p) \rangle = \langle (1\ x), (1\ x\ \dots) \rangle = S_p.$$

Note:  $\langle (1\ 3), (1\ 2\ 3\ 4) \rangle$  cannot generate  $S_4$ . Observe that

$$(1\ 3)^2 = \text{id}, \quad (1\ 2\ 3\ 4)^2 = (1\ 3)(2\ 4), \quad (1\ 2\ 3\ 4)^3 = (1\ 4\ 3\ 2)$$

$$(1\ 3)(1\ 2\ 3\ 4) = (1\ 2)(3\ 4)$$

$$(1\ 3)(1\ 2\ 3\ 4)^2 = (2\ 4)$$

$$(1\ 3)(1\ 2\ 3\ 4)^3 = (1\ 4)(2\ 3)$$

$$\text{So, } \langle (1\ 3), (1\ 2\ 3\ 4) \rangle = \langle (1\ 2)(3\ 4), (1\ 3)(2\ 4) \rangle.$$

Prop.  $A_4$  has no subgroup of order 6.

Proof. Let  $V = \{id, (12)(34), (13)(24), (14)(23)\}$  be a subset of  $A_4$ .

Except  $id \in V$ , the elements of  $V$  have order of 2, exactly, these are all elements of order 2 in  $A_4$ .

The Conjugate cannot change the order and parity,  $V$  is a normal subgroup of  $A_4$ .

And, since  $A_4/V$  has order of 3, it must be contain order 3 element, so, it must be isomorphic to  $\mathbb{Z}_3$ , cyclic.

Meanwhile, if there exists a order of 6 subgroup,  $N$ , then since  $[A_4 : N] = 2$ , thus for any  $g \in A_4$ ,  $gN = Ng$ . so it is normal.

Now,  $A_4/N \cong \mathbb{Z}_2$ , cyclic.

Meanwhile, observe that: for  $G' = \{x^{-1}y^{-1}xy \mid x, y \in G\}$ , and  $N \trianglelefteq G$ ,

$G/N$  is abelian iff  $G' \leq N$

To show this, let  $x^{-1}y^{-1}xy \in G'$  be given. Since  $G/N$  is abelian,

$$x^{-1}y^{-1}xyN = x^{-1}N y^{-1}N xN yN = x^{-1}N xN y^{-1}N yN = N$$

so  $x^{-1}y^{-1}xy \in N$ .

Therefore,  $G' \leq N$  and  $G' \leq V$ . And, since  $A_4$  is not abelian,  $G' \neq 1$ .

And, if  $|G'| = 2$ , then it must be contain an order of 2 element, but since  $G'$  is normal, then it has all order of 2 element.

Thus,  $G' \geq V$ , or  $G' = V$ . Now,  $|G'| = 4$  but  $G' \leq N$ , it is contradiction to the Lagrange Thm.